The Incumbency Paradox in Indian Elections

Checkpoint 1 – Project Status Report

Team 043 **Mentor:** Mayank Jain

Section 1: Meta Information

1. **Open Project Title:** The Incumbency Paradox in Indian Elections – An Analysis of Win Rates and Margins
2. **Group Number:** 043
3. **Mentor Name:** Mayank Jain
4. **Member Names and Roll Numbers:** Ram (25b1079), Arjun (25b0963), Ruthvik (25b0981), Praveen (25b1037).
5. **GitHub Repository Link:** The Incumbency Paradox in Indian Elections

Section 2: Progress So Far

2.1 Descriptive Statistics of the Dataset(s)

Used the TCPD Lok Dhaba dataset to analyse Indian Parliamentary elections. The source dataset contains approximately **91,669 records and 44 columns**. The dataset is processed and files are maintained in the GitHub repository.

2.2 Data Cleaning Steps Completed

Retained Lok Sabha general elections and excluded by-elections, which fill vacancies rather than represent a full general-election cycle. Incumbent and non-incumbent records were separated into CSV files, and data validation was completed. Year-wise winning rates and differences were calculated and saved in a summary CSV file. Also generated line graphs showing winning-rate trends and differences, along with histograms of winning margins.

2.3 Division and Revision of Responsibilities Within the Group

Ram (25b1079): Statistical tests and initial statistical analysis.

Arjun (25b0963): Close-contest preparation, Regression analysis.

Ruthvik (25b0981): Dataset collection, merging and cleaning using python and other tools

Praveen (25b1037): Winning-margin distributions, correlation between them.

2.4 Fine-Grained Problem Statement Refinement

The central problem is to understand whether candidates who have previously held office enjoy an electoral advantage in Indian Parliamentary elections compared with candidates who have not. Although incumbents may benefit from prior experience, public recognition, and established voter support, their observed electoral success may also reflect differences in candidate quality and political party selection. It is therefore important to distinguish the apparent advantage of incumbency from other factors that may influence election outcomes. The study also acknowledges that party-level factors may play a role in electoral success, although the current analysis primarily focuses on individual incumbency.

2.5 Updated Tracking Spreadsheet

The existing tracking spreadsheet has **not been modified** for this checkpoint.
SpreadSheet Link: SpreadSheet

2.6 Blockers/Challenges

Data validation for the current stage is complete. There are no critical blockers currently requiring immediate attention or guidance from the Teaching Team.

Next Steps:

Identify close contests using previous-election margins, establish a party-incumbency measure, and conduct the planned statistical analysis.